

Magnifying Augmented Mirrors for Accurate Alignment Tasks

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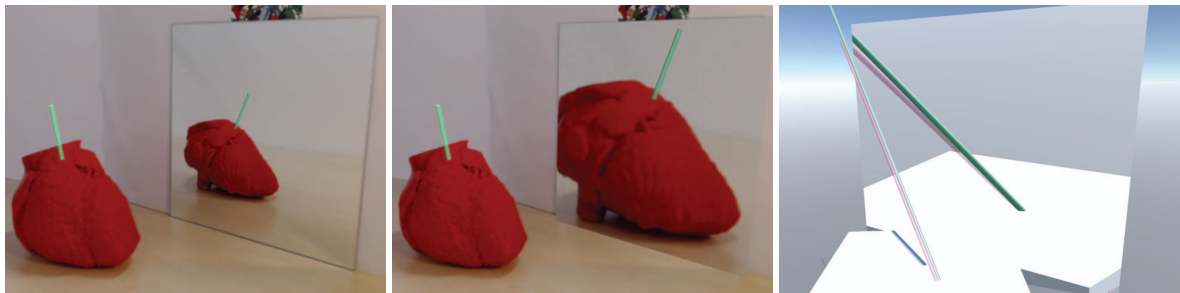


Figure 1: Augmented Mirrors enable the visualization of real and virtual content using real mirrors (left). Our method facilitates the observation of further details by providing a magnified view of the real and virtual content of the scene (middle). We conducted a virtual reality study in which users had to align a rod (green) to a reference (pink) with the help of a magnified mirror (right).

ABSTRACT

Limited mobility in augmented reality applications restricts spatial understanding along with augmentation placement and visibility. Systems can counteract by providing perspectives by tracking and augmenting mirrors without requiring user movement. However, the decreased visual size of mirrored objects reduces accuracy for precision tasks. We propose Magnifying Augmented Mirrors: digitally zoomed mirror images mapped back onto their surface, producing magnified reflections. In a user study ($N = 14$) conducted in virtual reality, we evaluated our method on a precision alignment task. Although participants needed time for acclimatization, they achieved the most accurate results using a magnified mirror.

Keywords: Human-centered computing, Human computer interaction (HCI), Interaction paradigms Mixed / augmented reality, Visualization, Augmented reality, Virtual reality

1 INTRODUCTION

Through stereopsis, the two human eyes create a visual cue to perceive the environment in three dimensions. Although head-mounted displays for virtual reality (VR) and augmented reality (AR) use stereoscopic renderings, it is difficult to precisely estimate the depth of an object from an egocentric view due to the small stereo base

between the eyes. Integrating non-egocentric viewpoints like co-located mirrors can help understand the spatial relationship of objects and the user's perspective. When positioned correctly, mirror reflections create a new perspective of an object of interest that reveals otherwise unseen areas and reduces the number of uncertain degrees of freedom, such as depth. This principle has been shown beneficial for alignment tasks using an AR tracked planar mirror [2]. However, objects appear smaller within the reflected image compared to the direct view as the perceived distance of objects increases. This could lead to safety hazards in medical or industrial AR if instruments are aligned based on a non-magnified mirror view. Restrictions in space can prevent the user from freely approaching the mirror for closer observation, which could lead to evaluation mistakes. As high accuracy is mandatory in industrial or medical contexts, too small reflections should be avoided. Therefore, we explore the magnification of the mirror reflection to allow users to perceive even small details within the reflection. Our AR approach uses a real mirror for coping with depth mismatches and applies magnification on the mirror image to offer a closer view. To achieve this, an additional camera is assessed, which records the mirror image from the user's point of view. The obtained image is magnified and mapped back onto the mirror surface, creating a magnified reflection that includes virtual objects and real surroundings. A medical application of our method is depicted in Figure 1. We evaluated our approach in a VR study with 14 participants and showed that the participants achieved more accurate results in the magnified mirror condition than when using an augmented mirror without magnification or no mirror at all.

2 RELATED WORK

The mirror paradigm can be used to cope with challenging depth mismatches, especially in medical AR. Martin-Gomez et al. [2] used a real mirror to provide a secondary perspective, which reflects both virtual augmentations and real surroundings. It helps to cope with the difficult depth perception, understand the scene, display

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Table 1: Alignment accuracy in rotation and distance. The arithmetic mean (AM) and the standard deviation (SD) are computed for the rotation (Rot) and distance (Dist) using no mirror (NoM), a mirror (Mir), and a magnifying mirror (MagM).

	NoM Rot [deg]	Mir Rot [deg]	MagM Rot [deg]
AM	4.96	1.64	1.08
SD	8.67	2.23	1.48
	NoM Dist [cm]	Mir Dist [cm]	MagM Dist [cm]
AM	0.31	0.70	0.61
SD	0.20	1.74	1.23

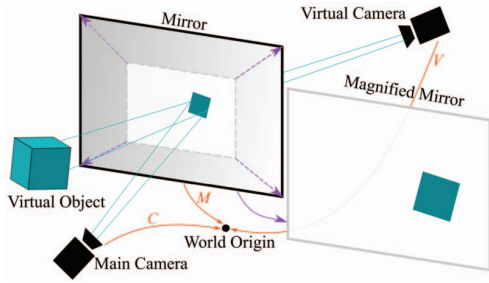


Figure 2: The position of the virtual camera can be determined with the transformations to C (main camera), M (mirror), and V (virtual camera) relative to the coordinate system from the origin. For the user study, the mirror image is cropped by a factor of 3 and interpolated back onto the mirror plane to create the magnification.

possible scenarios, and reveal occluded content. For alignment tasks, Dastan et al. [1] proposed an augmented collimation widget for assisting users in positioning drills in dental implantology with precision alignment. They show current angular and positional errors, which assists users with visual feedback about the current alignment. However, limited space can prevent the user from moving for a closer view and perceiving small details of augmentations [2, 3]. Different works address this problem by enlarging regions of interest in both AR and VR applications. Yu et al. [4] introduced magnification in a teleconsultation tool between AR and VR users to improve collaboration. Elements can be moved, rotated, scaled, and thereby magnified, including annotations to improve task precision. Therefore, we extend the magnification concept to help users in AR with restricted space to allow a closer observation without moving.

3 METHODOLOGY

Our approach is based on previous work from Martin-Gomez et al. [2], providing an implementation of a real mirror that reflects virtual augmentations. This work presents an additional magnifying function for the mirror image. We use digital zoom to magnify, cropping the mirror texture via a surface shader and interpolating the texture back onto the mirror plane. Textures for the augmentations and the background containing the real surroundings are magnified separately. The mirrored augmentations are drawn on top of the background to prevent occluding the augmentations. The setup and implementation of the magnification are depicted in Figure 2.

To test our solution's accuracy in a reproducible setup and mitigate the observation of external factors such as light or calibration of the AR device, we conducted the user study in a VR setup with a virtual mirror. Equally to the AR scenario, we set up the mirror plane and the additional virtual camera. The users can enable the magnification via a VR controller button. This way, accuracy for alignment tasks can be evaluated easily by comparing the aligned with a reference object.

4 PRELIMINARY EVALUATION

We hypothesize that magnification helps the users to perform the alignment task, increasing their accuracy. Magnification provides a closer view and should help the participants to make accurate judgments about their alignment quality.

$N = 14$ participants ($M = 25.5$, $SD = 2.59$ years, six males, eight females) participated in our user study. After a short familiarization phase, the participants were asked to align a virtual rod attached to a hand-held VR controller to a reference rod with either no mirror, a mirror, or a magnified mirror in randomized order. We measured the accuracy of the alignment by determining the angle and the closest distance between the reference and the user-held object, and collected their preferred method via two qualitative questions.

As denoted in Table 1, the best combinations of distance and orientation were achieved with the magnified mirror. With the non-magnified mirror, the distance was larger, and the orientation matched less closely. Although the closest distance measure was smaller using no mirror relative to the other two tasks, the angle between the rods was larger. This means that the two had one close point of almost intersection, but otherwise, the corresponding points on the rods are diverging, making the alignment most inaccurate. Using a mirror the orientation matched more closely, meaning that even with a higher distance all remaining points on the rods are closer to each other and, therefore, more accurately aligned.

We asked the participants about their preferred method, and ten of them favored the method with a non-magnified mirror due to the additional global perspective of the scene. They claimed that the mirror provided realistic and easy support requiring less cognitive load. We assume that people are already familiar with regular mirrors from daily life and, therefore, adopt the method faster.

However, ten participants thought they achieved the most accurate alignment with the magnified mirror due to the closer view and the better visibility of smaller differences and details. Some participants stated that longer training time would help with the magnified mirror.

5 DISCUSSION AND CONCLUSION

We assume that magnification helps with executing tasks more accurately. Our user study has shown that in an alignment process, the participants could achieve more accurate results with a magnified mirror. Nevertheless, the users need some time for acclimatization. In the future, we plan to conduct a larger user study. We want to improve our magnification method by implementing an optical zoom for the mirror plane to provide a more realistic magnification by accessing and minimizing the camera's field of view. This would also maintain the initial resolution of the mirror image.

In this poster, we proposed a magnification method for augmented mirrors. The promising results of the user study lead us to believe that this method can be a useful tool in precise alignment tasks, and we invite the community to further explore methods of integrating magnified secondary perspectives into AR scenes.

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